Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
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Project Title:	Spaceflight Holography Investigation in a Virtual Apparatus (SHIVA)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
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Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:

The program objective will be met through the following tasks: 1. Apply and refine the theoretical analysis to help specify the flight experiment matrix. (completed) 2. Conduct computer experiments to assist with the hardware requirements and specification. 3. Collect and analyze additional data from the IML-1 holograms. (completed) 4. Produce a preliminary experiment design. (completed) 5. Specify the required measurement capability. (completed) 6. Breadboard and test the experiment. (completed) 7. Perform ground feasibility demonstration. (completed) 8. Specify requirements of a flight package. (completed) 9. Conduct digital holography experiments to establish the data extraction and interpretation procedures for video data. (completed) 10. Satisfy the recommendations of the SCR panel. (completed) a. Conduct parametric experiments to test theoretical predictions over a wider range. b. Conduct particle interaction studies, c. Conduct calibration experiments. d. Identify and work with an experiment statistician to optimize the proposed flight experiment matrix 11. Support MSFC Engineering Design and prototyping. (completed) 12. Reviews and Reports: i) Science Requirements Document (SRD) – First Draft Completed. ii) Science Concept Review (SCR) – Completed. iii) Requirements Definition Review (RDR). The objectives of the experiment are to: 1. Test and validate new solutions of the general equation of motion of a particle in a fluid. 2. Collect data that will help extend existing solutions: Particle interaction, Wall effects, Higher Reynolds number, Non spherical, Crystal moving in a fluid. 3. Characterize the ability of isolation tables to isolate experiments containing particles in liquids. 4. Characterize the low frequency acceleration environment of the Space Station. 5. Explore applications of digital holography and related techniques such as electron speckle pattern interferometry (ESPI) and laser doppler velocimetry (LDV) to monitoring the health of spacecraft. The movement of a particle in a fluid is one of the most fundamental processes in physics and plays an important role in materials science. For example, the physics of crystal growth from a nucleation site in solution can be dominated by movement of the solution over the crystal surface. Gravity and the Stokes drag dominate the particle equation of motion under normal gravity conditions, but in the absence of a strong gravitational field, other transient forces can become relevant. The consideration of these forces makes the equation of motion much more complex. Until recently, the equations were solved only by numerical methods and/or by neglecting terms. During this study, we discovered an exact solution to the equations, which shows that the usually neglected terms become extremely important in microgravity. We also developed diagnostic recording methods using holography to save all of the particle field data, allowing the experiment to essentially be transferred from space back to earth in what we call the "virtual apparatus".
The program was removed from Flight status during the first quarter of 2004 and redirected to be more compliant with the “New Space Vision”. Despite the cancellation of the spaceflight experiment, most of the objectives of the program have been achieved because the ground experimental data proved good enough to verify the theoretical particle equation of motion over most of the conditions intended for the spaceflight experiment. The most important condition of the spaceflight experiment that cannot be duplicated in ground experiments is the examination of particle motion as the Reynolds number is varied from low to high. During the past year, the program has been redirected toward applying the optical technique used in the ground experiment (digital holography) to new challenges such as detecting defects in spacecraft components. In addition to digital holography, other related laser-based optical techniques are being investigated, such as electronic speckle pattern interferometry (ESPI) and laser doppler velocimetry (LDV).

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

1. Parametric experiments to test theoretical predictions over a wider range The derivation of the particle equation of motion assumes certain simplifications: 1) the particle is spherical, 2) the particle does not interact with walls or other particles, and 3) the particle Reynolds number is low. (The particle Reynolds number is proportional to the particle radius times its velocity relative to the fluid divided by the kinematic viscosity of the fluid.) There are many practical situations where these simplifications are not valid. We conducted several tests to examine the limits of validity of the theoretical equation of motion. The motion of tethered particles in the ground experiment is highly repetitive, and we took advantage of this by applying strobe methods to capture sequences of images of the moving particle at small phase increments of the oscillation cycle. We completed a series of ground experiments using the strobe method where particles at four different densities were shaken over a range of frequencies. The resulting image sequences were digitally reconstructed and animated to produce movies of the moving particles. The particle-cell amplitude ratio was determined by comparing their trajectories. In addition, the movies made plainly visible the phase difference between the particle and cell motions, which were measured to a fraction of a degree. Since the phase difference and amplitude ratio are both predicted by the particle equation of motion, strobe measurements provide a rigorous test of its validity. The experimental results were in excellent agreement with the theoretical predictions for spherical particles at Reynolds numbers < 5.
Our experiments with non-spherical crystals yielded responses that were less than for an equivalent volume spherical particle. We also conducted a number of particle interaction experiments, where particles were placed very close to walls or two particles were placed close to each other. None of these experiments showed any clear repeatable effect of interaction.

An important difference between the ground experiments and the planned space-flight experiments is that fine wires (tethers) were required to anchor the particles in place during the ground experiments. The tethers would not have been needed in the space-flight experiments. The tethers cause the particle to describe an arc rather than a straight line as the fluid oscillates. For low oscillation amplitudes, the deviation from linear motion is insignificant, thus the particle equation of motion could be experimentally verified at low Reynolds numbers. However, to study particle motion at Reynolds numbers > 5 would have required oscillating the cell at large amplitudes where the deviation from linear motion would have been too large to ignore. The arc deviation can be reduced by using a longer tether, however, this creates other problems: a larger cell is needed to accommodate the longer tether, and more powerful shaker is needed to move the larger cell. Furthermore, a longer tether should experience larger tether drag, which is difficult to correct for. The conclusion is that the tethered particle method is poorly suited for examining particle response over a large range of Reynolds numbers.

The effect of particle Reynolds number on the response of particles is the most important question that we have not been able to address in the SHIVA study. Further experiments are needed to derive and verify a more comprehensive particle equation of motion that includes particle shape and particle Reynolds number. We have not found a satisfactory method for examining Reynolds number effects in ground-based laboratories. If it ever becomes possible to do this experiment in microgravity, a systematic investigation of the Reynolds number effect (i.e., from $0 < Rep < 20$) should be the highest priority. Also our ground experiments allow us to conclude that a much simpler and less costly spaceflight experiment

Task Progress:	could be conducted using a smaller cell with many particles under investigation simultaneously. 2. Spacecraft Health Monitoring by Digital Holography and Laser Doppler Velocimetry SHIVA produced some extremely valuable experiments and theory that have applications in NASA's new space initiative. We are exploring two concepts: holography in a virtual apparatus, and spacecraft health monitoring by laser Doppler vibrometry. The transmission of three-dimensional views across interplanetary distances will play a major role in future space missions. Holography is a logical choice for recording three-dimensional information. To be practical for space applications, holograms must be recorded and transmitted digitally, then reconstructed by computer at an earth-bound location. This is the essence of what we attempted to achieve in the SHIVA program: record holograms of particle fields in an experiment aboard the International Space Station, transmit the holograms to earth, reconstruct the holograms by computer, and then make detailed optical measurements on the reconstructed images. During the ground experimental program, we successfully demonstrated that we could digitally record holograms and reconstruct the focused images, and that we could obtain precise and accurate optical measurements. To be truly useful for space exploration, however, digital holography must be developed much further than the current state of the art. The chief obstacle in using electronic media to record holograms is that they are not able to resolve the very fine fringe spacings resulting from most types of illumination. The fringe spacing decreases as the angle between the object and reference beams increases. The experimental system that we chose to study during the SHIVA program consisted of large opaque particles in a transparent fluid. Such a system is amenable to in-line holography, in which a narrow field of view is illuminated by an in-line laser source. At present, the resolving power of the best electronic recording media is about 100 times less than that of the best film media, so digital holography is impractical if the angle between object and reference beams is greater than a few degrees. We anticipate that future technology will produce sensors with greater numbers of pixels and higher pixel densities, which will make digital holography much more practical. The development of laser sources is also important for the future of digital holography. The main requirement for laser sources used in holography is that the beam coherence length must be longer than the distance between the object and sensor. This implies that the laser bandwidth must be fairly narrow. High power is also important, and pulsed lasing followed by gated detection is useful for recording holograms under conditions where the ambient light levels are high. For space applications, the lasers must be small, light-weight, robust, and consume little power. Fiber lasers developed for telecommunications are particularly promising sources for holographic applications in space. MetroLaser is a world leader in the development of laser instruments for measuring vibration. One of the instruments that we have developed is a single point laser Doppler vibrometer – the MetroLaser Vibromet 500. Throughout the SHIVA program, this instrument proved to be invaluable for evaluating the ground and spaceflight hardware. We anticipate that Laser Doppler Vibrometry (LDV) will also prove valuable during the design and construction of future spacecraft and for monitoring the structural integrity of these spacecraft. We performed a preliminary experiment with a NASA-provided test panel consisting of nine shuttle tiles to determine if LDV could effectively detect bonding defects. The detection principle is that the tile will be freer to vibrate when the bonding to the underlayer is weak. Since the Vibromet 500 is a single-point measuring system, we divided the test surface into a 1 inch square grid pattern, and made measurements at the center of each grid square. Each tile required 11 x 11 separate measurement trials. In each trial the sample was acoustically excited, then the surface vibration at the grid point was monitored by a Vibromet 500. In the preliminary experiment, we found that the LDV method could detect most types of defects. design.

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